

ROCCO DI TELLA

rocco.di_tella@cfa.harvard.edu

EDUCATION

Universidad de Buenos Aires (UBA)

2020 - 2024

Licenciatura in Data Science (Joint B.S. + M.Sc.)

GPA: 9.0/10

Thesis Advisors: Enzo Tagliazucchi, Gabriel Kreiman

Thesis project: Convolutional Neural Networks as models of human visual attention.

RESEARCH EXPERIENCE

Predoctoral Fellow — AstroAI, Harvard Center for Astrophysics

sept 2026 - present

Advisor: Cecilia Garraffo. SAO Predoctoral Fellowship.

Generative AI architectures that respect known physical symmetries.

Postbaccalaureate Fellow — SPARK, Harvard Center for Astrophysics

sept 2025 - aug 2026

Advisor: Juan Rafael Martínez Galarza.

Simulation-based inference to correct photon pile-up in X-ray observations of black-hole accretion.

Visiting Student — AstroAI, Harvard Center for Astrophysics

june - aug 2025

Advisors: Cecilia Garraffo, Juan Rafael Martínez Galarza. SAO Visiting Student Fellowship.

Flow matching with transformers to generate multi-modal posteriors for galaxy images and X-ray spectra.

Research Assistant — Applied Artificial Intelligence Lab, UBA

jan - june 2025

Advisor: Luciana Ferrer

Automatic detection of hallucinations in generative language models using grey-box calibration methods and white-box probing. Continued as a remote collaborator through Dec 2025, supported in part by the AISAR AI Safety Scholarship.

Research Assistant — Kreiman Lab, Harvard Medical School

2024 (remote)

Advisor: Gabriel Kreiman

Licenciatura thesis co-advisor; collaboration continued after graduation.

Convolutional neural networks as models of human visual attention.

Research Assistant — Center for A.I. and Neuroscience, UTDT

2021 - 2022

Developed Bayesian methods for efficient experiment design. Conducted data analysis and designed experiments in studies of human cooperation.

Summer Research Intern — A.I. and Robotics Lab (LINAR), Universidad de San Andrés 2021

Deep learning for flight control automation.

ACADEMIC PUBLICATIONS

- **Di Tella R.**, Cuesta-Lazaro C., Böttger RL, Reeves J., Martínez-Galarza R.
“Probabilistic probes for galaxy evolution: signatures of AGN feedback in the AION foundation model”, *NeurIPS 2026 Workshop on Interpretability for Discovery*
- Gago J.*, **Di Tella R.***, Shi J.
“Thinking out of distribution: what a PhD must train in the age of AI”, *AJE Advances: Research in Epidemiology* (2026, in press)
- **Di Tella R.***, Böttger RL*, Cuesta-Lazaro C., Cappelluti N., Reeves J., Martínez-Galarza R.
“Exploring the connection between optical and X-ray”, *Conference on Physics and AI (PAI26)*, Stanford University (2026)
- Pincirolì N., Di Tella R., Cuesta-Lázaro C, Smith MJ, Garraffo C, Martínez-Galarza R.
“Augmenting Representations with Scientific Papers”, *Neural Computing and Applications* (2026)
- Borrell T., Dillmann S., Duraphe K., Eris F., Iyer K., Khederlarian A., Kumar A., Marraffini G., Smith MJ, Sourav S., Di Tella R., Wu JF
“The Platonic Universe: Do Foundation Models See the Same Sky?”, *ICML 2026 Workshop on Mechanistic Interpretability*; arXiv:2509.19453

- Pérez-Carrasco MI, Park CF, Zhu Q, Di Tella R, Ayazpour Z, Abad GG, Garraffo C
“Reconstruction Is Not Enough: Evaluating Self-Supervised Objectives for Spectroscopic Satellite Embeddings”, *ICLR 2026 Workshop on Machine Learning for Remote Sensing*
- Rieznik A., Di Tella R., Schwartzman L., Babino A. (2021)
“Optimum Integration Procedure for Connectionist and Dynamic Field Equations”, *Frontiers in Neuro-robotics*
- Cavanna F., Pallavicini C., Milano V., Cuiule J., Di Tella R., González P., Tagliazucchi E. (2021)
“Lifetime Use of Psychedelics is Associated With Better Mental Health Indicators During the COVID-19 Pandemic”, *Journal of Psychedelic Studies*

*Equal contribution.

TALKS & POSTERS

- “Exploring the connection between optical and X-ray emissions in Active Galactic Nuclei”, AstroAI Seminar, Center for Astrophysics, 2026 (talk).
- Tutorial on agentic AI for researchers, AstroAI Workshop, CfA, 2026 (invited tutorial speaker).
- “The Platonic Universe: Do Foundation Models See the Same Sky?”, AstroAI Workshop, CfA, 2026 (poster and short talk).
- “The Platonic Universe: Do Foundation Models See the Same Sky?”, New England Mechanistic Interpretability Workshop (NEMI), 2026 (poster).
- “Augmenting Representations with Scientific Papers”, ICLR 2026 Workshop on Foundation Models for Science (poster).
- “Where’s Waldo? How Human Brains and Neural Networks Look for Stuff Differently”, AstroAI Seminar, Center for Astrophysics, 2025 (talk).
- “Modeling X-ray photon pile-up with a normalizing flow”, NeurIPS, 2025 (poster).
- “Building Honest Agents through introspection”, AstroAI workshop, 2025 (poster).
- “Finding Waldo”, Simposio Científico de Inteligencia Artificial y Aplicaciones, 2024 (poster).
- “A Bayesian Look”, Bayes Plurinacional Workshop, 2024 (short talk and poster).
- “Fast Nonparametric Bayesian Framework for On-the-Fly Adaptive Design Optimization”, Bayes Plurinacional Workshop, 2023 (poster).
- “A Massive Experiment on Choice Blindness”, Sociedad Argentina de Neurociencia, 2017 (poster).

TEACHING EXPERIENCE

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| Lead Instructor — Fundación Sadosky | <i>aug - sept 2026</i> |
| <ul style="list-style-type: none"> • Designed and taught the applied statistics module of <i>Fundamentals of Data Science and AI for the Energy Industry</i>: theory lectures, exercises, and coordination of three instructors leading the practical sessions | |
| Teaching Assistant — Universidad de Buenos Aires (UBA) | <i>aug 2025</i> |
| <ul style="list-style-type: none"> • Intensive winter course on applications of AI for astrophysics | |
| Teaching Fellow — Universidad de San Andres (UdeSA) | <i>june - july 2025</i> |
| <ul style="list-style-type: none"> • Advanced undergraduate course on Machine Learning | |
| Teaching Assistant — Universidad de Buenos Aires (UBA) | <i>jan - dec 2024</i> |

- Helped launch two new graduate courses: Bayesian Inference (taught half of the theory lectures) and Causal Inference
- Advanced undergraduate course on Statistics

Teaching Assistant — Universidad Torcuato Di Tella (UTDT)

2022 and 2024

- Undergraduate course on Cognitive Science
- Undergraduate course on Statistics

FELLOWSHIPS & SCHOLARSHIPS

SPARK Postbaccalaureate Fellowship — Center for Astrophysics, Harvard & Smithsonian 2025
Twelve months of funded full-time research.

Visiting Student Fellowship — Smithsonian Astrophysical Observatory 2025
Ten-week research visit at AstroAI.

AI Safety Scholarship — AISAR (AI Safety Argentina) 2025
Six-month research stipend and compute.

PROGRAMS & SERVICE

CAMBRIA — Cambridge Boston Alignment Initiative 2026
Three-week program on mechanistic interpretability and alignment.

Reviewer — Workshop on Representations for the Physical Sciences 2026

INDUSTRY & OUTREACH

Editorial Board Member — Ciencia Hoy nov 2025 - present
Popular-science magazine of the Universidad Nacional de Tres de Febrero (UNTREF).

Data Scientist — GRIDX Biotech Venture Capital 2021 - 2022
Started a data consulting team which grew into an independent company. Financial forecasting and consulting for portfolio companies.

Data Scientist — El Gato y La Caja 2017 - 2021
Engaged in data storytelling aimed at a general audience and designed experiments for cognitive science research. Click [here](#) to see showcased work.

Science Communicator — Instituto Baikal 2019 - 2020
Coordinated “Frontiers of Science”, a seminar series for a general audience.

SKILLS

Methods: simulation-based inference, normalizing flows and flow matching, probing and sparse autoencoders, Bayesian and causal inference, agentic AI workflows for research
Tools: Python (PyTorch, JAX), R, Julia